

Survivor Detection Challenge

Predicting Passenger Survival with a Denoised Feature Ensemble Pipeline

SURVIVOR_DETECTION_CHALLENGE | DATA CONSULTANT SUBMISSION

Challenge: Survivor Detection Challenge — Maritime Classification (Modified)

Abstract

Predicting individual survival outcomes in disaster scenarios is a classic benchmark for demonstrating a complete, disciplined classification workflow under noisy, incomplete data conditions. This project builds a supervised classification pipeline that predicts **Outcome** — whether a passenger survived a maritime disaster — using a dataset deliberately modified with injected Gaussian noise, decoy columns, and realistic missingness in age and cabin records. Following exploratory analysis of missingness patterns, category-wise survival rates, and correlations, the raw noisy columns were denoised and combined with engineered features (title, family composition, fare-per-person, cabin availability, and group size) into a clean 31-feature model-ready dataset. Three base classifiers — Random Forest, Histogram Gradient Boosting, and Logistic Regression — were compared individually and as a soft-voting ensemble using Repeated 5-Fold Cross-Validation. The ensemble achieved a mean validation accuracy of **0.8226** with the lowest variance across folds, and final predictions were validated for format and distributional consistency before submission.

1. Introduction

1.1 Problem Statement

Determining which passengers are likely to survive a disaster event has long served as a foundational tabular classification problem, valuable for illustrating disciplined data cleaning, feature engineering, and model selection practices. The task is framed as a binary classification problem: given demographic, family, and ticketing attributes of a passenger, predict the binary variable Outcome. The dataset reflects realistic adversarial data conditions — injected Gaussian noise across numeric fields, decoy columns outside the official schema, and missing values in age and cabin records — all of which must be identified and corrected before reliable modeling can begin.

1.2 Objectives

- Profile the dataset to identify noise artifacts, decoy columns, missingness patterns, and target balance.
- Recover clean categorical and numeric signal from noisy raw columns (title, class tier, family counts).
- Engineer family, fare, and cabin-availability features that capture non-obvious survival signal.
- Train and compare multiple classifiers using repeated cross-validation.
- Select a final model based on both accuracy and stability across folds.
- Generate a correctly formatted, distributionally consistent submission file.

2. Dataset Description

The dataset consists of individual passenger records from a maritime disaster, split into a training set of 712 rows and a test set of 179 rows, with 16 and 15 columns respectively. Table 1 summarizes the officially documented schema.

Column	Description
PassengerName	Full passenger name, including embedded title
TicketTier	Passenger class proxy — continuous, rounds to 1 / 2 / 3
Gender	Passenger sex
Age	Passenger age — 27.7% missing in train
RelativesAboard	Siblings / spouses aboard — noisy, rounds to a non-negative integer
ParentsChildren	Parents / children aboard — noisy, rounds to a non-negative integer
TicketCost	Fare paid — noisy, contains negative values
Berth	Cabin identifier — 77.7% missing in train
BoardingPort	Port of embarkation
Singleton	Flag for travelling alone
FarePerPerson	Fare divided by family group size
Title	Provided honorific field
Outcome (Target)	0 = did not survive, 1 = survived

Table 1. Dataset schema and feature descriptions.

A preliminary diagnostic pass — row counts, data types, missingness percentages, and duplicate checks — confirmed a clean load with zero duplicate passenger names, and surfaced three columns outside the official data dictionary: **PassengerId** and **TicketCost** containing implausible negative values, and **Class**, a 558-unique-value string column resembling raw ticket numbers, treated as a decoy rather than a usable feature. The target class balance was 62.4% did-not-survive versus 37.6% survived.

3. Methodology

3.1 Exploratory Data Analysis

EDA covered missing-value patterns, category-wise survival rates, and numeric distributions by outcome. Survival rate was examined across Gender, TicketTier, BoardingPort, extracted Title, and Singleton status, with the expected pattern emerging: female passengers and those in higher ticket tiers survived at substantially higher rates. Age and log-transformed fare distributions were compared between survivors and non-survivors, and family size was found to have a non-monotonic relationship with survival — very small and very large family groups survived at lower rates than small-to-mid-sized groups.

Two missingness-linked signals stood out: passengers with a **known Age** survived at a notably higher rate (40.8%) than those with missing Age (29.4%), and passengers with a **recorded Berth value** survived at more than double the rate (64.8%) of those without one (29.8%) — indicating that Berth availability functions as a strong proxy for cabin class and socioeconomic status rather than as a feature to be discarded solely because of its high missingness.

3.2 Denoising and Feature Engineering

Rather than using the raw noisy columns directly, each was reconstructed into a clean form before modeling, summarized in Table 2.

Feature	Description
Title	Extracted from PassengerName; rare titles bucketed into 'Rare'
Male	Gender, reconciled/back-filled using extracted Title where missing
Tier	TicketTier rounded and clipped to the valid 1–3 range
Sib / Par	Family counts rounded and clipped to non-negative integers
FamilySize / Alone	Derived from Sib + Par + 1; binary flag for travelling alone
IsChild / Mother	Age- and Title-based flags for children and travelling mothers
LogFare / LogFarePP	Log-transformed fare and fare-per-person to reduce right-skew
TicketGroup / SurnameGroup	Group-size features capturing shared tickets and family units
HasBerth	Binary flag for a recorded cabin value — a strong survival proxy

Table 2. Denoised and engineered features used in modeling.

Train and test sets were concatenated prior to feature construction so that group-based features (ticket group size, surname group size) were computed consistently across both, then split back apart afterward. The resulting feature matrix contained 31 columns for both the 712-row training set and the 179-row test set, with zero remaining missing values.

3.3 Model Training and Validation

Three base classifiers were trained and compared: a Random Forest (500 trees, minimum leaf size 2, sqrt max features) with median imputation, a Histogram Gradient Boosting classifier (learning rate 0.05, max depth 3, L2 regularization 1.0), and a Logistic Regression (C = 0.5) with median imputation and standard scaling. A soft-voting ensemble combining all three was evaluated alongside the individual models. Model performance was assessed using Repeated Stratified 5-Fold Cross-Validation with 3 repeats, scored on accuracy.

4. Results and Model Performance

The four candidate models achieved the following cross-validated performance, summarized below.

Selected Model	Mean CV Accuracy
Soft-Voting Ensemble	0.8226 ± 0.0239

Figure summary — final validation performance of the selected model.

Model	Mean Accuracy	Std. Deviation
Random Forest	0.8211	± 0.0292
Histogram Gradient Boosting	0.8188	± 0.0291
Logistic Regression	0.8239	± 0.0309
Ensemble (Soft Voting)	0.8226	± 0.0239

Table 3. Repeated 5-fold cross-validation accuracy by model (3 repeats).

Logistic Regression achieves the highest mean accuracy in isolation, but its fold-to-fold standard deviation is also the largest of the four candidates. The soft-voting ensemble trades a small amount of peak accuracy for the **lowest variance across folds**, making it the more reliable choice for a single final submission where consistency matters as much as peak score.

EDA findings carried through into the final feature set: **HasBerth** and the engineered **Title** categories were expected to rank among the most informative predictors, consistent with the raw survival-rate gaps observed for cabin availability and gender/title during exploratory analysis. Family-derived features (FamilySize, Alone, TicketGroup) contributed additional non-linear signal that a tree-based model can exploit more effectively than a purely linear classifier.

5. Conclusion

This project demonstrates a complete, disciplined classification pipeline: the dataset was profiled for noise artifacts, decoy columns, and missingness; noisy raw columns were denoised and combined with domain-informed engineered features covering title, family composition, fare, and cabin availability; and three base classifiers were compared individually and as a soft-voting ensemble via repeated cross-validation. The result — a mean validation accuracy of 0.8226 with the lowest variance among the candidate models — represents a reliable and well-calibrated final submission. The predicted survival rate on the test set (39.7%) closely tracked the training survival rate (37.6%), confirming no material prediction bias.

6. Future Work

- Hyperparameter tuning via grid or Bayesian search across all three base models.
- Stacking (a meta-learner) in place of soft voting, to potentially exceed the current ensemble accuracy.
- Deeper feature interactions, e.g. Title × Tier or FamilySize × Fare, to capture compound effects.

- Model-based (iterative or KNN) imputation for Age in place of simple median imputation.
- Threshold tuning if the competition scoring metric weights precision and recall asymmetrically.

Appendix — Technical Stack

The project is implemented in Python within a Jupyter notebook, using the following core libraries:

- pandas & numpy — data loading, cleaning, and feature computation
- matplotlib — exploratory and diagnostic visualizations
- scikit-learn — imputation, scaling, pipeline construction, cross-validation, ensemble modeling, and metrics

Key output artifacts generated by the pipeline:

- submission.csv — final Outcome predictions in the required submission format
- Survival-rate bar charts by category, age/fare distribution plots, and correlation-with-target visualization